CIS-1 Assignment 2

Write Your Own MD Engine, then Compare NAMD and OpenMM on a Protein

Deadline: Monday, 16 November 2026, 11:59 PM (IST). Individual work. Late submissions are not accepted.

This assignment follows the LJ class (where you wrote an MD code for argon) and Tutorial 2 (NVE simulation of Trp-cage in NAMD and OpenMM). Everything you need was covered there. Use the same commands and files as in those tutorials.

1 What you do

Part	Task	Marks
A	Write your own MD engine for your protein in its water box and check it against OpenMM	30
B	Run the **same NVE simulation in NAMD and in OpenMM** and compare the energies over time	35
C	Compare the three engines (yours, OpenMM, NAMD)	20
D	Write about what your protein does in nature	10
	Neatness of report and code	5
	Total	**100**

We have already prepared the protein files for you (the `.prmtop` topology and a ready-to-run `.rst7` starting structure, with water, ions and the box). You do not need to build the system yourself.

2 Your protein

Your protein (a 4-letter PDB ID) will be released separately. Find its page on the RCSB Protein Data Bank, `https://www.rcsb.org`, by searching for the ID. From that page, download the structure in the **legacy PDB format** (`.pdb`) using the *Download Files* menu. Open it in VMD or any viewer, and use the page for Part D.

3 One system, one set of settings

In **all** parts (your engine, OpenMM, NAMD) you simulate the **same system**: your protein in its water box with ions, in a periodic box, at constant energy (NVE). So that the three engines are comparable, they all use the same simple settings:

- periodic box, with the *minimum image* idea: each atom interacts with the nearest copy of every other atom;
- non-bonded interactions (van der Waals and Coulomb) cut off sharply at **9 Å** (no PME, no switching, no long-range corrections);
- all bonds flexible (no constraints), timestep **1 fs** unless stated otherwise;
- no thermostat and no barostat; initial velocities random at 300 K.

The `.prmtop` file contains every parameter. The two short articles at the end of this document (Appendix A: periodic boundaries, Appendix B: non-bonded interactions) explain the box and the cutoff.

Starting point. For all three engines, start from the same prepared files for your protein (`.prmtop` and `.rst7`). They are already minimised and equilibrated, so go **directly to the production NVE run**: no minimisation, no heating, no equilibration steps of your own.

4 Part A: your own MD engine for your protein (30 marks)

You will write an MD code for **your assigned protein in its water box**. We provide the topology, a reader that gives you plain arrays (masses, charges, bond/angle/dihedral lists and parameters, which atoms to skip in the non-bonded sum), a helper for angles and dihedrals, a neighbour-list helper so that the code is fast enough, a starter file, and a checker that compares your code with OpenMM. The energy is the sum of five terms: bonds, angles, dihedrals, Lennard-Jones and Coulomb. The LJ part is the same idea as in the LJ class; the other terms are new.

1. **(12)** Complete the starter file: energy and forces for each of the five terms, including the periodic box and the 9 Å cutoff. The checker compares each term with OpenMM. Show that it passes.
2. **(8)** Add the velocity Verlet loop and run 2 ps of NVE at 1 fs. Plot kinetic, potential and total energy against time and report how much the total energy changes.
3. **(5)** Make a table of the five energy terms at the starting structure. Which are largest, and which are positive or negative? Why?
4. **(5)** Run with timesteps of 0.5, 1 and 2 fs for the same simulated time. Plot how much the total energy fluctuates against the timestep. What do you see, and why?

5 Part B: NAMD and OpenMM on your protein (35 marks)

Use the files for your protein from the provided folder, and the settings above. Run NAMD and OpenMM **one after the other**, not together.

1. **(20) NVE run.** Run both engines for 2 ps of equilibration plus 10 ps of production at 1 fs. For each engine report the average temperature, average potential energy, how much the total energy fluctuates, and whether it drifts. Plot kinetic, potential and total energy against time. Explain why kinetic and potential energy mirror each other, and why the temperature still fluctuates although nothing controls it.
2. **(10) Timestep.** Repeat for 0.5 fs and 2 fs (short runs of 2 ps are enough). Tabulate the energy fluctuation for each timestep and engine. Which timesteps work well and which do not?
3. **(5) Look at it.** Show one picture of your protein in its water box (VMD) and plot how far the backbone moves from its starting structure over time. Does the protein stay folded?

If a run is too slow on your laptop you may shorten production, but say so in the report.

6 Part C: comparing the three engines (20 marks)

1. **(8)** For the starting structure, make one table of bond, angle, dihedral, van der Waals, electrostatic and total potential energy from all three engines (NAMD and OpenMM in kcal/mol; convert if needed). Which terms agree in all three and which differ? Suggest a reason for each difference.
2. **(7)** Compare the 2 ps NVE runs of the three engines: how much does the total energy fluctuate or drift in each? Which is best and why might they differ? (A sharp cutoff makes the energy drift; explain why.)
3. **(5)** Write a short conclusion (half a page): how far can you trust each engine, what did your own code get right or wrong, and what does energy conservation in NVE tell you?

7 Part D: what does your protein do? (10 marks)

Using the RCSB page and UniProt (cite your sources), write about one page in your own words:

1. What the protein does in nature and which organism it comes from.
2. Its shape: mainly helices, sheets or both, how many residues, and whether the structure came from X-ray or NMR.
3. Why scientists use it in simulations, or what makes it interesting.
4. How the simulation relates to this: its total charge, how many ions were added, and whether 25 ps is long enough to see anything biological.

Copied text scores zero.

8 Submission

Submit one zip named `rollno_PDBID.zip` by 16 Nov 2026, 11:59 PM, containing:

- `report.pdf` (up to 10 pages) with your plots, tables, and answers, and Part D as its own section. Label all axes with units.
- Your code for Part A and the log and output files from Part B.
- A note saying clearly if you used any tool or AI help. You must understand everything you submit; a short viva may follow.

9 Tips

- Start early: the runs take time, especially on a laptop, and your own code is slowest.
- NAMD energies are in kcal/mol and OpenMM energies in kJ/mol (1 kcal = 4.184 kJ). Convert before comparing.
- Do not minimise before NVE with random velocities; use the prepared files as they are.
- Copied code or results score zero for the whole assignment.

A Short article: periodic boundary conditions

Why a box? A simulation can only hold a few thousand atoms, but a real liquid is huge. If we simply put our protein and water molecules in a vacuum ball, the molecules at the surface behave very differently from those inside, and the water would evaporate. *Periodic boundary conditions* (PBC) remove the surface: the simulation box is imagined to be repeated infinitely in all directions, like tiles on a floor. Every atom therefore has neighbours on all sides, as inside a bulk liquid.

What happens at the edge. If an atom leaves the box through the right face, its periodic image enters through the left face with the same velocity. The number of atoms in the box never changes, and the total momentum is conserved. In the code, you may wrap a coordinate back into the box ($x \to x - L\,\mathrm{floor}(x/L)$), but forces must always be computed from the correct nearest copy (below).

Minimum image convention. Atom i could interact with atom j and with all of its infinite periodic copies. We keep only the *nearest* copy. For a rectangular box with side lengths L_x, L_y, L_z the displacement along each axis is

$$\Delta x \;\leftarrow\; \Delta x - L_x\,\mathrm{round}\!\left(\frac{\Delta x}{L_x}\right),$$

and the same for y and z. After this, every component satisfies $-L/2 \le \Delta x \le L/2$, and the distance is $r = \sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2}$. Always apply this to $\mathbf{r}_i - \mathbf{r}_j$ *before* computing r, forces and energies.

Condition for this to work. Each atom must see at most one copy of any other atom. This needs the cutoff r_c to be smaller than half of the shortest box side, $r_c < L_{\min}/2$. For a 9 Å cutoff the box must be at least 18 Å in every direction; the prepared boxes satisfy this.

Things to remember.

- Atoms of one molecule that sit on opposite faces of the box are not "broken": in the topology they are still bonded. Compute bonded terms with the minimum-image displacement too, or keep molecules whole.
- Unwrapped positions (never wrapped back) are what you need to follow a molecule's motion over time; wrapped positions are what you see in a viewer. Use whole molecules when you compute RMSD.
- NAMD reads the box from its configuration (cell vectors); the `.rst7` file also stores it. Use the same box in every engine.
- NVE keeps the box fixed; there is no barostat.

B Short article: non-bonded interactions

In an Amber-type force field, atoms interact in two ways. *Bonded* terms hold the molecule together: bonds $k(r - r_0)^2$, angles $k(\theta - \theta_0)^2$ and dihedrals $k[1 + \cos(n\phi - \delta)]$. *Non-bonded* terms act between atoms that are not close neighbours in the molecule: van der Waals (Lennard-Jones) and electrostatic (Coulomb). They are the most expensive part of an MD step because every atom interacts with many others.

Lennard-Jones (van der Waals). Two neutral atoms repel strongly when they overlap (electron clouds) and attract weakly at larger distance (induced dipoles):

$$u_{\mathrm{LJ}}(r) = 4\varepsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r}\right)^{12} - \left(\frac{\sigma_{ij}}{r}\right)^{6}\right], \qquad \sigma_{ij} = \frac{\sigma_i + \sigma_j}{2}, \quad \varepsilon_{ij} = \sqrt{\varepsilon_i \varepsilon_j}.$$

This is exactly the LJ class potential. The second rule is the *mixing rule*: the force field stores one σ and ε per atom type, and the pair values are built from them.

Coulomb (electrostatics). Each atom carries a partial charge q_i (in units of the proton charge). The energy of a pair is

$$u_{\mathrm{C}}(r) = \frac{C\, q_i q_j}{r}, \qquad C = 332.0637 \text{ kcal Å mol}^{-1}\text{e}^{-2}.$$

It falls off only as $1/r$, so it is long ranged. Real simulations use Particle Mesh Ewald (PME) to include all periodic copies correctly (used in Tutorial 2). In this assignment we use the simplest alternative, described next.

Cutoff. We ignore all pairs with $r > r_c = 9$ Å. Then the cost per step grows only linearly with the number of atoms (each atom has a fixed number of neighbours inside r_c). Practical points:

- Use r^2 and compare with r_c^2; avoid square roots until needed.
- A *neighbour list* (all pairs within r_c plus a small buffer, rebuilt every few steps) avoids testing all N^2 pairs at every step. The provided helper builds it for you.
- A sharp cutoff makes the energy jump when a pair crosses r_c. This adds small errors and a slow energy drift in NVE, more for Coulomb than for LJ because Coulomb is still large at 9 Å. This is one of the things to discuss in Part C.

Exclusions and 1-4 scaling. Atoms that are directly bonded (1-2), or form an angle (1-3), already interact through the bonded terms, so their non-bonded interaction is *excluded* (left out). Atoms separated by exactly three bonds (1-4 pairs, the end atoms of a dihedral) interact, but with

scaled strength: in Amber the Coulomb energy is divided by 1.2 and the LJ energy by 2.0. Atoms further apart (four or more bonds, or in different molecules) interact fully. The topology provides these lists; getting them right is the most common source of a wrong total energy.

Water. The water model here (TIP3P) is made only of bond, angle, LJ (on oxygen) and Coulomb terms. Since all bonds are flexible in this assignment, water is treated like any other molecule.

Energy units and checks. Energies are in kcal/mol, distances in Å, charges in e. A good check: the total force on the system must be zero (Newton's third law), and the energy terms must agree with a trusted engine at the same structure.